Predictive Fault Detection in Robots Using Deep Learning

Project Proposal

Project Advisor:

Mam Sarah Khaleel

Group Members:

Asjal Abdullah 22L-6273

Muhammad Waleed 22L-7788

National University Of Computer and Emerging Sciences
Department of Computer Science Lahore,
Pakistan

Abstract

Robots are widely used in industries, autonomous vehicles, and space exploration. One big challenge is that robots can fail unexpectedly due to problems in their motors, battery, or other internal systems. Most existing methods detect these faults only after they happen, which can cause delays, damages, or accidents.This project aims to predict robot failures **before** they happen using deep learning. A simulated robot will operate in a simple environment, and its internal data—such as speed, acceleration, battery level, motor load, and task time—will be recorded. Using this historical data, a deep learning model called **LSTM** (Long Short-Term Memory) or **GRU** (Gated Recurrent Unit) will learn patterns that happen before a failure occurs.The model will output a fault risk percentage for the robot at each step. For example, it can warn: "Battery low, motor load high, predicted risk 80%, reduce speed or stop robot." This approach allows the robot to take action **before a failure happens**, making it more reliable.This project is fully robotics-related because it focuses on robot health and predictive maintenance, not just vision or navigation. It combines robotics simulation with deep learning and gives clear, measurable results that can be tested in Python.

1. Introduction

Robots are increasingly used in industries, autonomous vehicles, and automated systems to carry out important tasks. Despite their design for reliability, robots can still experience unexpected failures due to issues such as motor overload, low battery, or other internal system problems. Traditional fault detection methods usually identify these problems **after** they occur, which can lead to downtime, equipment damage, safety hazards, and higher costs. This reactive approach is inefficient, especially in critical robotics applications. To overcome this, the project focuses on **predictive fault detection**, which aims to anticipate failures before they happen. A simulated robot will operate in a Python-based environment, performing tasks while tracking internal parameters such as speed, acceleration, battery level, motor load, and task duration. A deep learning model, like LSTM or GRU, will be trained on this historical data to identify patterns that indicate an imminent failure. The model will provide a risk score at each step, enabling the system to suggest preventive actions such as slowing down or stopping the robot. This proactive method enhances robot reliability, safety, and efficiency, making it highly relevant to real-world robotics applications where preventing failures is crucial.

2. Divison of Tasks

To ensure efficient project development, the tasks will be divided among the group members based on their responsibilities in simulation development, data processing, and deep learning implementation.

1. Asjal Abdullah (22L-6273)

- Design and implementation of the **Python-based robot simulation environment**.
- Development of modules to generate and record robot parameters such as speed, acceleration, battery level, motor load, and task duration**.**

- Data preprocessing and preparation of time-series datasets for training.
- Integration of the simulation system with the predictive model for testing.
- Contribution to system evaluation and preparation of technical documentation.

2. Muhammad Waleed (22L-7788)

- Research and selection of appropriate **deep learning models (LSTM / GRU)** for predictive fault detection.
- Implementation and training of the neural network using the collected dataset.
- Model optimization, tuning, and performance evaluation using metrics such as accuracy, precision, and recall.
- Assistance in Development of the fault risk prediction module that outputs the probability of robot failure.
- Assistance in testing the complete system and preparing the final report and presentation.

Joint Responsibilities

Both group members will work together on several parts of the project. We will jointly write and review the **Python code** used for the robot simulation and the deep learning model. Both members will help in **testing the code**, fixing errors, and improving the overall system performance.

We will also work together to **connect the simulation with the prediction model** so that the robot data can be used to predict possible faults. During development, we will discuss the results and make improvements where needed.

In addition, both members will contribute **to** analyzing the dataset, checking model results, and validating whether the predictions are accurate. Finally, we will jointly prepare the project report, documentation, and presentation slides for submission and final evaluation.

References

[1] https://www.kaggle.com/datasets/shivamb/machine-predictive-maintenance-classification

[2] Forecasting failure rate of IoT devices: A deep learning way to predictive maintenance
https://dl.acm.org/doi/10.1016/j.compeleceng.2023.108829

[3] Machine Learning Approaches for Predictive Maintenance in IoT Devices By Mazedur Rahman 1, *, Amir Razaq 1, Md. Tanvir Hossain 2 and Md Towfiq Uz Zaman
https://wjaets.com/sites/default/files/fulltext_pdf/WJAETS-2025-1388.pdf